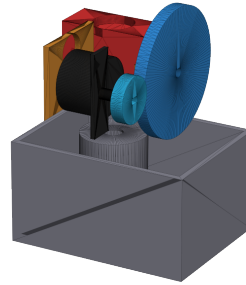


Pan/Tilt Design-Similarity Study

Comparing the original reference assembly to the current 28BYJ-48 geared redesign, and proposing ideas to make them look and feel more alike



Original: ASSEMBLY_pan_tilt.STEP



Current: pan_tilt_geared_28byj48

This is a PROPOSAL document. Nothing here has been built into a new CadQuery script yet - it lays out what the two designs actually look like today, why they diverged, and a menu of concrete options (with trade-offs) for pulling the redesign's look and topology closer to the original, without violating three hard constraints the user has set: keep the current ~90 x 85 x 117mm envelope, keep the 28BYJ-48 motor, and keep every part 3D-printable or off-the-shelf.

Chapter 1 - Original Design Overview

Mechanical_Design/ASSEMBLY_pan_tilt.STEP (also present as a .stp duplicate and a CATProduct) is a SolidWorks-2022-exported STEP AP203 assembly: a full-size, gear-driven camera pan/tilt head. It is not an original laser-pointer design - it predates this project and was adopted purely for its serial pan-tilt-stack topology (one motor pans the whole upper structure, a second motor carried on that structure tilts the payload), which matches the chosen mechanical concept (Concept A). Measured directly from the STEP file: 76 distinct solid bodies, overall bounding box **245 x 298 x 269 mm**, all 76 solids report as valid B-rep (a stricter shell-closure check flagged one part, the four-bolt flange bearing, in an earlier project session - both checks are worth trusting for what they each test, not treated as contradictory).

The STEP file carries no surviving PRODUCT/part names (SolidWorks did not export them, or the export flattened the tree) - the identifications below are by shape and size, matched where possible against the loose .CATPart files also sitting in Mechanical_Design/ (Motor base, Cam Holder, Gear 2 Base, Baring 2 shaft, Spacer, spur gears, etc.). Treat this as a confident visual read, not a verified bill of materials.

Visual character

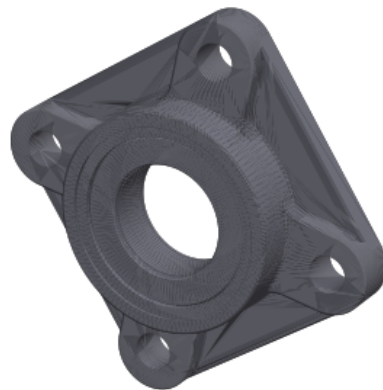
Three things dominate the original's look, in order of visual impact:

- **Bent/faceted bracket arms.** At least three large, near-identical fork-shaped brackets (~180-266mm long) with diagonal stiffening creases and bolt-hole patterns - a sheet-metal-bend or cast-and-machined look, not a boxy printed look.
- **Real gears.** The 60T and 20T spur gears have actual involute teeth cut (module 1.125, 20° pressure angle) - a strong, instantly recognisable silhouette cue that a placeholder disc does not give.
- **Visible off-the-shelf hardware.** A large four-bolt flange (pillow-block style) bearing, cylindrical wall ball bearings, a NEMA-style motor can with a square 4-bolt flange, and flat mounting plates with big lightening cutouts and diagonal ribs all read as "machined metal assembly", not "printed enclosure".

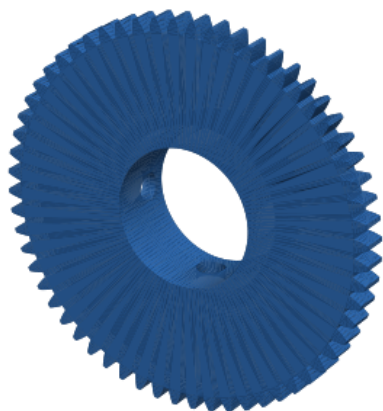
Representative parts



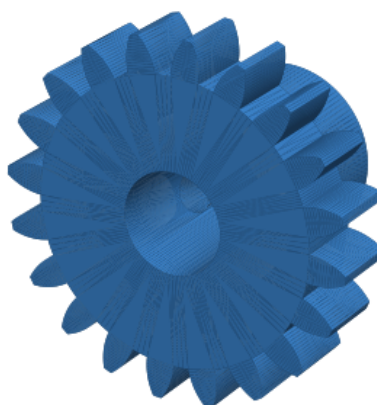
Fork/bracket arm - one of 3 forming the yoke structure



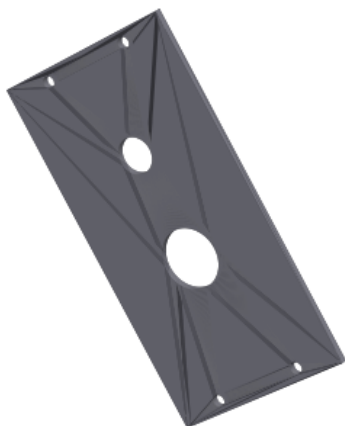
Four-bolt flange (pillow-block) bearing



60-tooth spur gear - real involute teeth



20-tooth pinion - real involute teeth



Flat mounting plate - lightening holes + diagonal ribs



Generic NEMA-style motor can, square 4-bolt flange
(already superseded by 28BYJ-48)

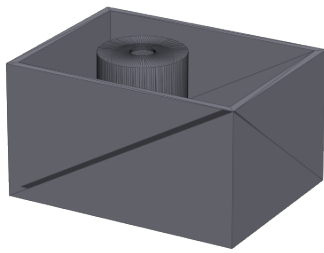
Chapter 2 - Current Geared Redesign Overview

Python scripts/pan_tilt_geared_28byj48.py → Mechanical_Design/pan_tilt_geared_28byj48.step is a from-scratch CadQuery rebuild sized natively around the 28BYJ-48 stepper, built after the original direct-drive attempt (pan_tilt_28byj48.py) was frozen due to unfixed geometry bugs. It keeps the original's serial pan-tilt-stack topology and its 3:1 external gear reduction on both axes (20T/60T, module 1.0, 20° PA - same tooth counts and pressure angle as the original, module scaled down for print fidelity), but everything else is redrawn: real 28BYJ-48 geometry (5mm D-shaft, 8mm shaft/can offset, "inboard-clocked" motor orientation), real small-format bearings (608ZZ pan / 625ZZ tilt) instead of the original's oversized bearings and invalid flange bearing, and steel shaft rod instead of printed shafts.

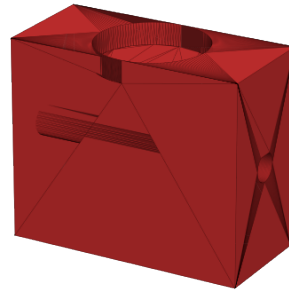
17 valid closed solids, structural envelope (base+platform+yoke) 90 x 68 x 104mm, full envelope including motors/gears 90 x 85 x 117mm. Every interference check in the script's own automated report reads 0.0 mm³ (including a ±90° yoke sweep), tilt-axis torque margin 19.6x, all-PLA mass budget ~333g total.

Visual character

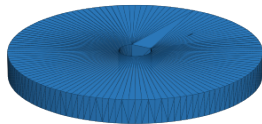
The redesign currently reads as "clean FDM enclosure": flat-walled boxes and rectangular blocks with 0.9mm corner reliefs (a deliberate FDM-friendly detail, not a style choice), plain cylindrical bearing pockets flush with the housing, and gear teeth not yet modelled at all - the gears are placeholder discs at the correct pitch diameter, correct for meshing/center-distance purposes but visually blank.



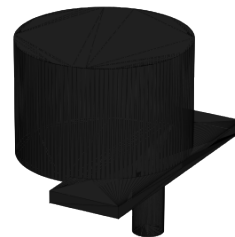
Base shell - plain box, no surface detail



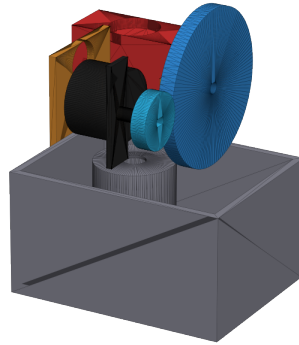
Yoke - plain rectangular block



Pan gear - placeholder disc, no teeth

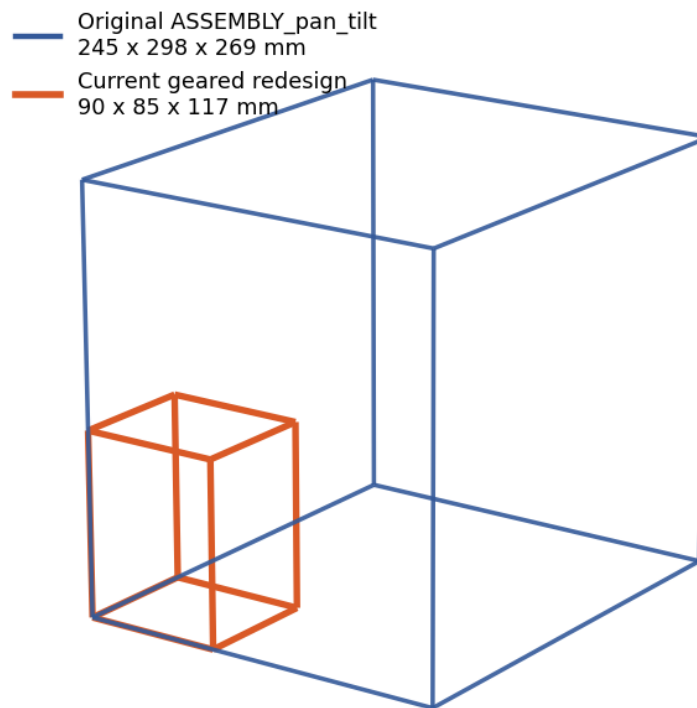


28BYJ-48 motor placeholder - can + 2-tab bracket



Full current assembly (base / platform / yoke / pan gear+pinion visible)

Chapter 3 - Side-by-Side Comparison



Bounding-box comparison, same scale - the redesign is roughly 2.7x smaller in every linear dimension (about 20x smaller by volume)

Aspect	Original ASSEMBLY_pan_tilt	Current geared redesign
Overall envelope	245 x 298 x 269 mm	90 x 85 x 117 mm (~2.7x smaller/axis)
Solids / parts	76 solids (STEP), ~17 unique per earlier BOM read	17 valid solids
Motor	Generic NEMA-style can, square 4-bolt flange	28BYJ-48, 2-tab bracket (fixed - not changing)
Gear stage	60T/20T, module 1.125, REAL involute teeth	60T/20T, module 1.0, placeholder discs (no teeth)
Reduction ratio	3:1 external per axis	3:1 external per axis (unchanged, matches original)
Bearings	4-bolt flange bearing (invalid B-rep) + wall ball bearings	608ZZ (pan) + 625ZZ (tilt), pressed, both valid
Structural style	Bent/faceted bracket arms, ribbed plates, lightening holes	Flat-walled printed boxes, corner reliefs, no surface detail
Shafts	Oversized (dia 25-40mm implied by bearing bores)	Steel rod, dia 8mm (pan) / dia 5mm (tilt)
Material language	Machined/bent metal + off-the-shelf steel hardware	3D-printed PLA + off-the-shelf small bearings

The one-line summary: topology (serial pan-tilt stack, 3:1 geared both axes, same tooth counts) is already a close match and does not need to change. What reads as "different" is almost entirely **surface language and part styling** - faceted/ribbed bent-metal brackets with real gear teeth and visible flange hardware, versus flat-walled printed boxes with blank gear discs. That is good news: most of the ideas in Chapter 4 are additive/cosmetic and do not require touching the envelope, the motor, or the already-validated interference-free geometry.

Chapter 4 - Proposed Ideas, by Subsystem

Each subsystem below lists 2 options with trade-offs. Effort/risk is rated against the already-validated current geometry (0 mm³ interference at every check, 19.6x tilt torque margin) - anything marked LOW risk is additive and should not require re-running those checks; anything marked MEDIUM/HIGH risk changes structural geometry and would need the checks re-run.

A. Gear stage - real teeth vs. placeholder discs

The single strongest visual cue in the original that the redesign is currently missing.

- **A1 - Cut real involute teeth** [MEDIUM effort, LOW-MEDIUM risk] - Generate actual 20T/60T involute profiles at the existing module 1.0, 20° PA, CD=40mm. Meshing is already governed purely by center-distance (verified by an in-code assertion), so this is a drop-in swap for the disc geometry - envelope, torque budget and interference checks are all unaffected in principle, though gear-vs-gear/gear-vs-housing clearance needs re-checking once real tooth tips exist. Also unlocks computing real backlash instead of quoting the source PDF's 0.3-0.5° estimate.
- **A2 - Keep placeholder discs** [NONE, LOWEST risk] - Zero new work. Correct for meshing/CD, but the biggest single reason the redesign currently reads as "unfinished" rather than "small version of the original".

Recommendation: A1. This was already flagged as the natural next step in prior sessions and is the highest-leverage single change for visual similarity - do it before any of the cosmetic options below.

B. Motor mount styling

The 28BYJ-48's real tab pattern (2 tabs, 35mm pitch) cannot be changed to match the original's 4-bolt square NEMA flange without lying about the motor's actual geometry.

- **B1 - Square-flange-styled bracket** [LOW effort, LOW risk (bracket is not load-bearing structure)] - Keep the real 2-tab fixing (accurate to the motor) but shape the bracket plate itself as a rounded square flange with 4 cosmetic corner bosses (2 real M4 clearance holes + 2 blind decorative bosses), echoing the original's square-flange language without pretending there are 4 real bolts.
- **B2 - Leave as-is** [NONE] - Current thin 2-tab plate is the lightest, most minimal, most "honest" option, but reads the least like the original of any subsystem.

Recommendation: B1 - cheap, zero part-count or motor-compatibility cost, and it is the piece most people will instinctively compare against the original's motor mount.

C. Base / pan-platform silhouette

The original's forked/faceted brackets and ribbed plates are its most photogenic feature; the redesign's base and platform are currently plain rectangular boxes.

- **C1 - Cosmetic ribs + lightening holes** [LOW effort, LOW risk] - Add diagonal surface ribs and circular lightening cutouts to the base shell and platform deck faces (surface treatment only, wall thickness and outer envelope untouched).
- **C2 - Forked/tapered platform walls** [MEDIUM-HIGH effort, MEDIUM risk] - Reshape the platform's tilt-supporting walls from flat rectangles into a tapered fork silhouette (wider at the base, narrowing toward the tilt axis) - the single most recognisable shape element of the original's bracket arms. Bigger CAD change; needs the tilt-bearing bore positions and wallA/wallB clearance checks re-verified.

Recommendation: Do C1 first (safe, immediate visual win). Offer C2 as a follow-up phase if the user wants a stronger silhouette match and is fine re-running the structural checks.

D. Yoke shape

The original's yoke IS its forked bracket arms; the redesign's yoke is a solid rectangular block (already validated: 0 mm³ interference across a full ±90° sweep, 19.6x torque margin).

- **D1 - Slim forked yoke** [MEDIUM-HIGH effort, MEDIUM risk] - Rework the yoke into a two-arm fork straddling the payload pocket, tapering toward the tilt shaft - visually much closer to the original, and would

reduce yoke mass (a bonus for the torque margin, not a requirement given the current 19.6x headroom). Requires wall-thickness/rib design for the fork arms and a full re-run of the sweep + torque checks.

- **D2 - Cosmetic scallops on the current block** [LOW effort, LOW risk] - Keep the validated solid block, add side pockets/scallops that suggest a forked look without changing the outer envelope or the bore/keying geometry at all.

Recommendation: D2 first - it is free relative to the current validated geometry. D1 is the more authentic match and worth doing eventually, but only as a deliberate follow-up phase, not bundled into the first similarity pass.

E. Bearing/shaft housing styling

The original's four-bolt flange bearing is a very recognisable "cast pillow block" look; the redesign's 608ZZ/625ZZ pockets are invisible, flush cylindrical bores.

- **E1 - Cosmetic flange boss around the pan bearing tower** [LOW effort, LOW risk] - Add a square flanged boss with 4 corner pads (decorative - the bearings are pressed, not bolted, so no functional bolts are needed) around the outside of the existing bearing tower. Purely an additive shell feature; does not touch the pocket bore or press-fit dimensions.
- **E2 - Leave bearing towers unadorned** [NONE] - Cleanest and lightest option, but the least visually similar of any subsystem to the original's most "machined-part"-looking component.

Recommendation: E1 - same cost/risk profile as B1 and C1, and closes out the last major "visible hardware" gap between the two designs.

F. Umbrella recommendation - surface language, not a scaled clone

At ~90mm the redesign cannot be a literal scaled-down clone of the 245-298mm original: the original's thin bent sheet-metal sections would either need proportionally much thicker walls to print reliably at this scale, or would be too thin to print at all. The practical reading of "similar" here is **family resemblance through applied surface language** - real gear teeth, ribs/lightening holes, a styled motor flange, a styled bearing boss - layered onto geometry that stays FDM-appropriate (the existing 0.9mm corner reliefs, wall thicknesses, and press-fit tolerances all stay as-is). This is what options A1/B1/C1/D2/E1 collectively deliver, at LOW-MEDIUM total risk to the already-validated model.

Chapter 5 - Recommendation & Next Steps

Suggested phase 1 (low risk, high visual payoff)

- A1 - real involute gear teeth (20T/60T, module 1.0, CD=40mm unchanged)
- B1 - square-flange-styled motor brackets (real 2-tab fixing preserved)
- C1 - cosmetic ribs + lightening holes on base/platform faces
- D2 - cosmetic scallops on the existing yoke block
- E1 - cosmetic flange boss around the pan bearing tower

All five are additive or drop-in swaps against the current, already-validated 90 x 85 x 117mm geometry - envelope and motor both untouched, and none of them should require anything beyond re-running the existing interference/torque checks as a final sanity pass.

Suggested phase 2 (bigger structural rework, only if phase 1 isn't enough)

- C2 - taper the platform walls into a forked silhouette
- D1 - rework the yoke into a slim two-arm fork

Open questions before writing any new CadQuery code

1. Does phase 1 (surface-language changes only) already satisfy what "very similar" means, or is the forked-arm silhouette (phase 2) important enough to do now rather than later?
2. OK to proceed with real involute gear teeth (A1)? It is real script work (new tooth-profile generation code) even though it shouldn't move the envelope or torque numbers.
3. Should phase-1 work stay strictly additive (never re-deriving the yoke sweep or torque checks from scratch, just re-confirming they still pass), or is it fine to also revisit those numbers if a phase-1 change happens to touch them?
4. Any preference on which subsystem to prototype first once you've reviewed this - gears (A1) are the highest visual leverage, but motor bracket (B1) or bearing boss (E1) are the fastest to turn around if you want a quick win to look at before committing to the gear-tooth work.

Source files: Mechanical_Design/ASSEMBLY_pan_tilt.STEP (original), Python scripts/pan_tilt_geared_28byj48.py → Mechanical_Design/pan_tilt_geared_28byj48.step (current). This report: Python scripts/generate_pantilt_similarity_report.py → Docs/PanTilt_Similarity_Ideas.pdf.